Jingkun Jiang

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Education

Hunan University, China Sep 2024 - Jun 2027
M.S. in Materials Science and Engineering Advisor: Prof. **Yang Xia**

- Scientific Computing, Phase-Field Modeling, High-Performance Computing (HPC)

Hebei University of Architecture, China Sep 2020 - Jun 2024
B.Eng. in Materials Science and Engineering **Rank 3/73 (5%)**

Publications & Patents

[1] **J. Jiang**, P. Deng, and Y. Xia. "sp-DBA: a general framework for adaptive transform-domain computation." Preprint, 2026. [arXiv:2609.03922]

[P1] **J. Jiang** and Y. Xia. "A GPU-Oriented Adaptive Acceleration Method Using Spectral Dynamic Block Activation in the Fourier Domain." Chinese Invention Patent Application, No. 202611350264.9, 2026.

Research Experience

sp-DBA Framework for Adaptive Fourier-Domain Computation — Ongoing

- **Co-designed sp-DBA with my advisor** and **independently implemented** the framework and solver interfaces. Developed hardware-aligned block activation and numerical stability controls for Fourier-domain updates, and evaluated update-stage and overall performance against matching full-update Regular solvers.
- Applied sp-DBA to PFC polycrystalline coarsening, multicomponent reaction-cross-diffusion pattern formation, and **controlled digital backpropagation (DBP)** calculations.
- Organized the implementation around a shared core and lightweight solver interfaces while retaining the existing governing equations and numerical schemes.

Computational Materials Modeling with Phase-Field Methods — Ongoing

- Developed a 3D **phase-field (PF)** solver for dendritic growth, and **PFC, XPFC, and APFC** solvers for **crystalline microstructure evolution** including grain coarsening, interface growth, and defect evolution.
- Implemented finite-difference and Fourier pseudo-spectral solvers in **C++**, using fourth-order exponential time-differencing (ETD4) schemes for PFC/XPFC simulations, and accelerated large-scale simulations using **OpenMP and MPI parallelization**.
- Implemented **density-peak extraction, local orientation estimation, grain segmentation,** and **grain-size distribution (GSD)** analysis for PFC simulation fields.

Distributed FFT and Multi-GPU Scaling — Ongoing

- Built a distributed 2D FFT using slab decomposition, local cuFFT transforms, MPI-based redistribution, and CUDA-aware communication to support distributed sp-DBA.
- Evaluated **strong and weak scaling on 1-8 NVIDIA A800 GPUs** and identified communication overhead in distributed FFT as the main performance bottleneck at larger GPU counts.

Skills

Programming	C/C++, Python, MATLAB
Parallel/HPC	CUDA, OpenMP, MPI, distributed FFTs, multi-GPU scaling
Software	ParaView (PF/PFC), OVITO, VASP (DFT), Materials Studio

Honors & Academic Service

Provincial First Prize, Chinese Mathematics Competitions for College Students 2022, 2023
Outstanding Graduate, Hebei University of Architecture 2024
Teaching Assistant, Hunan University 2025 - 2026
Student Affairs Assistant, Hunan University 2026 - Present